VOID·SPACE

3D Digital Asset Management & XR Publishing Platform

Platform Reference & Feature Convergence Document

Part A — Existing Platform (Baseline + Blockchain/IPFS by Default) | Part B — Proposed Next-Generation Platform (EON Reality–Inspired Feature Convergence)

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Classification	Internal / Confidential
Purpose	Reference basis for the next VOID·SPACE SRS revision — combines the existing platform (with blockchain/IPFS added by default) and a proposed, natively-built feature set inspired by EON Reality's platform pillars.

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How to Read This Document

This document is organized into two independent parts so each can be lifted directly into the next VOID·SPACE SRS revision without cross-editing.

- Part A — Existing Platform: everything VOID·SPACE already specifies in VS-SRS-1.0, restated with full architecture diagrams, plus a blockchain/IPFS storage layer added as a default (non-optional) part of the baseline — since this was requested with “no issues at all” to add by default.
- Part B — Proposed Platform: a natively-built VOID·SPACE feature set that draws on EON Reality's platform structure (Learn / Train / Perform / Achieve) purely as an inspiration checklist — this is NOT an integration with or dependency on EON Reality. Every feature listed is something VOID·SPACE would build itself, combined with the DAM + blockchain foundation from Part A.
- Each EON Reality feature is scored KEEP / ADAPT / SKIP against VOID·SPACE's identity as a 3D-asset-first DAM platform, with a one-line reason, so you can prune the SRS scope quickly.
- A 3D asset creation tooling section (Blender + Unity, both free) is included per your note about where models/game-ready assets actually get built before they enter the pipeline.

PART A — Existing Platform (Baseline + Blockchain/IPFS by Default)

A.1 Overview & Purpose

VOID·SPACE is a web-based 3D Digital Asset Management (DAM) platform that lets Creators upload, organize, and submit 3D assets for review, and lets Assessors evaluate and approve those assets for downstream use in XR training pipelines. The baseline product covers the full asset lifecycle: ingestion (manual upload or third-party import), AI-assisted classification and description via the Claude API, human-in-the-loop review and approval, in-browser 3D preview, and publishing of approved assets to an XR training platform. It exposes both a browser UI and a documented REST API.

In this revision, a blockchain/IPFS layer is folded into the baseline architecture by default — every approved asset is content-addressed on IPFS and registered on-chain for provenance, without requiring a separate integration decision later.

A.2 Current Feature Set

Functional requirement modules from VS-SRS-1.0, summarized:

Module	Capability	Priority
3.1 Authentication & Access Control	Email/password + Google SSO login; role-scoped JWT (Creator / Assessor); API-key tokens for machine clients.	High
3.2 Asset Upload & Management	Upload .glb/.gltf/.obj/.fbx/.stl/.blend up to 200MB; required Name/Category, optional Tags; draft/pending status; Creator-owned library view, filter, delete.	High
3.3 Review & Approval Workflow	Assessor queue of pending assets; approve/reject/request-revision with comment; in-app notification to Creator on status change.	High
3.4 3D Preview & Visualization	Live rotating WebGL thumbnails; full-screen orbit-controlled viewer; metadata panel (polycount, format, tags, status).	High
3.5 Third-Party Tool Integrations	Sketchfab & Poly Pizza search/import by license and polycount; headless Blender conversion/optimization; Meshy AI text/image-to-3D; async job polling.	Medium
3.6 AI-Assisted Content Intelligence	Claude API generates suggested tags/description/quality flags as a structured JSON contract; suggestions are clearly labelled, never auto-approve an asset.	High
3.7 XR Publishing	Approved-only publish to the downstream XR platform; returns an XR manifest reference and module URL; rejects non-approved assets (HTTP 409).	Medium
3.8 Developer / API Access	Versioned REST API (/api/v1/) for auth, assets, tools, jobs, publishing; in-app API reference; API-key rotation dashboard.	Medium

A.3 User Roles

Role	Description	Proficiency
Creator	Produces or sources 3D assets, uploads with metadata, tracks review status.	Moderate — familiar with tools like Blender

Role	Description	Proficiency
Assessor	Reviews submitted assets against quality/XR-readiness criteria; approves, rejects, or requests revisions.	Moderate–High — XR/3D pipeline expertise
External Developer	Consumes the REST API to integrate VOID·SPACE into external pipelines.	High — software engineering
System Administrator	Manages accounts, API keys, and platform configuration.	High

A.4 Current High-Level Architecture

The baseline system is a browser client talking to a REST API backend, backed by a relational store and binary asset storage, with async jobs fanning out to AI and third-party 3D tooling, and a publish path into the downstream XR platform.

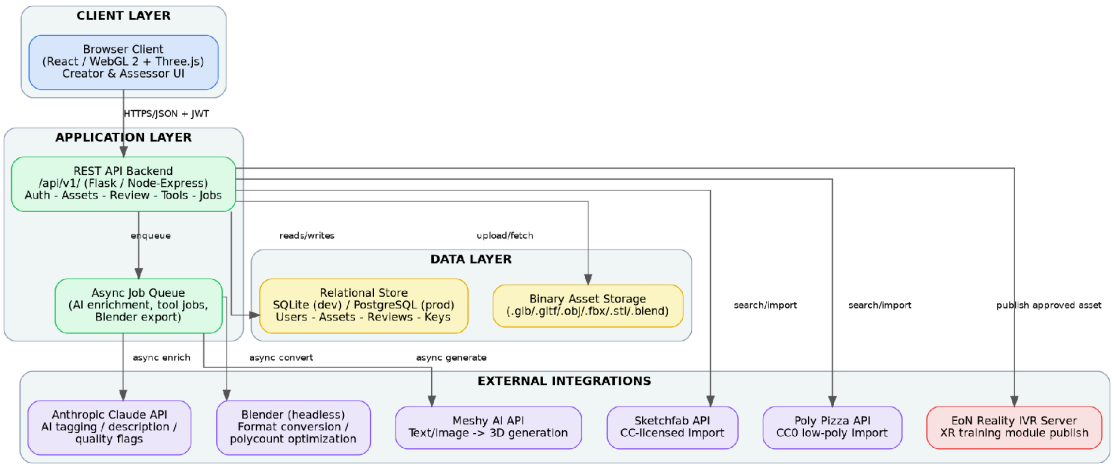


Figure A-1 — Existing VOID·SPACE architecture (baseline, pre-blockchain)

A.5 Blockchain / IPFS Layer — Added by Default

This is folded into the baseline rather than treated as an optional add-on. Two pieces are added to the existing stack:

- IPFS / Filecoin storage for asset blobs. On approval, the binary asset (not just its metadata) is pinned to IPFS via a pinning service (e.g. Pinata or web3.storage). The resulting Content Identifier (CID) is a hash of the content itself, so any later tampering with the file is immediately detectable — the CID simply won't match.
- A smart-contract asset registry on a low-fee L2 chain (e.g. Polygon), recording CID, content hash, owner/Creator address, license terms, and approval timestamp. This gives every approved asset a tamper-evident, independently verifiable provenance record that doesn't depend on VOID·SPACE's own database staying honest or online.
- Wallet connection (MetaMask / WalletConnect) is additive, not a replacement for the existing email/Google SSO auth — it's only invoked at the moment a Creator or Assessor needs to sign a blockchain transaction (e.g. confirming publish-to-chain).

> *Design intent: the relational database remains the system of record for day-to-day app state (fast reads, filtering, search). The chain + IPFS pair is the system of provenance — slower, but tamper-evident and independent of VOID·SPACE's own servers. The app should degrade gracefully if the chain or IPFS pinning service is temporarily unavailable, consistent with the existing NFR-REL.1 graceful-degradation principle.*

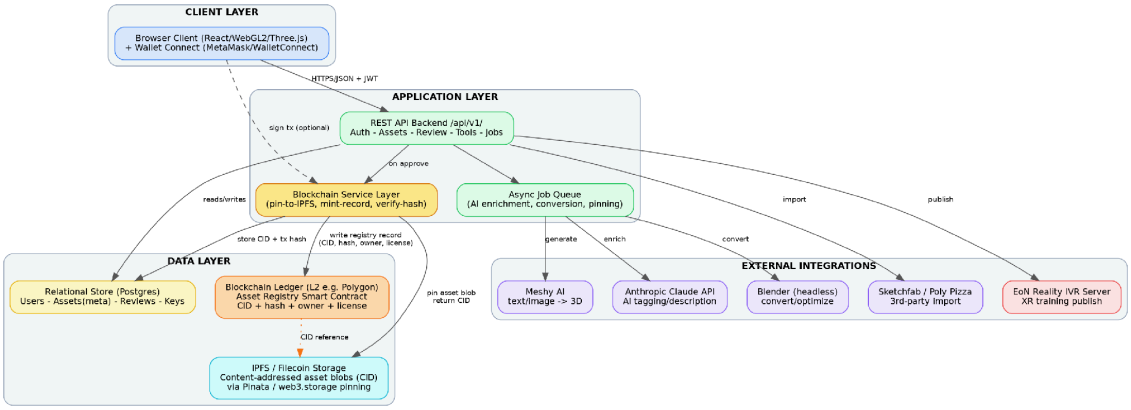


Figure A-2 — Existing architecture with the blockchain/IPFS layer added by default

A.5.1 What gets written on-chain vs. off-chain

Data	Where it lives	Why
Raw 3D asset file	IPFS (content-addressed)	Large binary; chain storage is prohibitively expensive per byte.
CID + content hash + owner + license + timestamp	On-chain registry contract	Small, needs tamper-evidence and public verifiability.
Searchable metadata, tags, review comments, status	Relational DB (existing)	Needs fast filtered queries; not something that benefits from immutability.
AI-suggested tags/description	Relational DB (existing, unchanged)	Editable/overridable by design — must never be immutable.

A.6 Technology Stack Summary

Layer	Technology
Client	React + WebGL 2 / Three.js, evergreen browsers
Backend	Python 3.11+/Flask or Node.js/Express, behind Nginx
Database	SQLite (dev) / PostgreSQL (prod)
Binary storage	Local/object storage (baseline) — see A.5 for IPFS addition
AI content intelligence	Anthropic Claude API (Messages endpoint)
3D tooling	Blender (headless conversion), Meshy AI (generative 3D)
3rd-party asset sourcing	Sketchfab API, Poly Pizza API
XR publishing target	EoN Reality IVR Server API
Blockchain (added by default)	EVM-compatible L2 (e.g. Polygon) + IPFS/Filecoin pinning

A.7 Non-Functional Highlights

- Performance: sub-400ms P95 for asset list queries up to 10,000 assets; thumbnails initialize within 1s.
- Security: salted password hashing, per-route bearer-token scope enforcement, upload validation by extension/size, no PII sent to the Claude API beyond filename/category/tags.

- Reliability: core upload/review flow keeps working if Claude, Sketchfab, EoN Reality, or (new) the chain/IPFS pinning service is unavailable — same graceful-degradation principle extends to the blockchain layer.
- Compliance: imported third-party assets retain their source license in metadata; on-chain license terms give this a durable, independently-checkable record.

PART B — Proposed Platform (EON Reality–Inspired Feature Convergence)

B.1 Vision & Positioning

VOID·SPACE is not integrating with EON Reality. EON Reality's public feature breakdown is used here purely as a checklist of proven XR-training-platform capabilities — each one is re-scoped as something VOID·SPACE would design and build natively, on top of its own DAM + blockchain foundation from Part A. Where a feature doesn't fit a DAM-first product, it's marked SKIP below rather than force-fitted.

The resulting product story: VOID·SPACE stays a 3D Digital Asset Management platform at its core, but grows an AI content-generation layer, a lightweight training/assessment layer, and a blockchain-backed provenance and marketplace layer around that core — so a single ingested document or asset can become a certified, monetizable, verifiable piece of training content.

B.2 Feature Adoption Matrix — EON Reality Pillars vs. VOID·SPACE

KEEP = build natively into VOID·SPACE as a core feature. ADAPT = build a lighter/DAM-appropriate version. SKIP = out of scope — not aligned with VOID·SPACE's product identity or overkill for a DAM-centric platform.

LEARN pillar

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
EON XR (no-code AR/VR authoring, shared 3D rooms)	No-code lesson authoring layered on top of the existing asset library, so an Assessor-approved asset can be dropped into a simple AR/VR lesson template without a 3D engine.	**KEEP**	Directly extends the DAM's core value — approved assets need somewhere to be used.
EON AI (auto curriculum/quiz/in-lesson guidance)	Reuse the existing Claude API integration (already generating tags/descriptions) and extend its prompt to also produce a lesson outline and quiz questions from the same ingested document.	**KEEP**	Same AI pipeline already in the baseline — this is a prompt/schema extension, not new infrastructure.
Sentient Worlds (persistent always-on 3D environments)	Not built as a standalone product.	**SKIP**	Persistent multiplayer world infrastructure is a large, ongoing operational commitment unrelated to asset management; revisit only if VOID·SPACE pivots toward a full LMS.

TRAIN pillar

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
Genesis Trainer (doc/SOP -> narrated 3D lesson)	Extend the ingestion pipeline: a document or SOP upload triggers AI generation of a narrated walkthrough that references existing 3D assets from the library.	KEEP	Natural extension of ingestion + AI content intelligence already in the baseline SRS.
Genesis Simulator (Awareness->Practice->Apply->Mastery scored loop)	A lightweight 4-stage progress tracker on top of the lesson/quiz data, not a full simulation engine.	ADAPT	The scoring/progression concept is useful; a full physics-based simulator is out of scope for a DAM platform.
Brainy Soft Skills (AI-avatar roleplay)	Not built.	SKIP	Conversational AI-avatar roleplay is a distinct product category (needs voice AI, avatar rendering) — high cost for low overlap with asset management.

PERFORM pillar

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
Assess IQ (AI-driven skills testing & compliance validation)	Reuse the existing Assessor review workflow's scoring concept; extend it into a formal AI-assisted quiz/exam scored against the ingested content, with results written to the on-chain registry as a verified-skill record.	KEEP	Pairs naturally with the existing human-in-the-loop review model and the new blockchain layer.
Assist IQ (always-on real-time AI co-pilot during real tasks)	Not built as always-on; a scoped “AI Mentor” chat inside a lesson view (see below) covers the core need.	ADAPT	Full real-time task co-piloting needs live sensor/task context VOID·SPACE doesn't currently ingest.
Field IQ (AR smart-glasses step-by-step field guidance)	Not built.	SKIP	Requires dedicated smart-glasses hardware partnerships and safety-critical certification — outside a DAM platform's scope.

ACHIEVE pillar

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
Career Compass (AI maps verified skills to jobs)	Not built as a full job-matching engine; the on-chain verified-skill record (from Assess IQ-equivalent above) is exportable/shareable so a learner can present it elsewhere.	ADAPT	Issuing a portable, verifiable credential is on-brand for a blockchain-backed DAM; building an actual job marketplace is not.
Venture Builder (AI mentor to launch a micro-business)	Not built.	SKIP	Business-mentorship product is unrelated to 3D asset management.

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
AI Fluency (structured AI-skills course path)	Not built as a dedicated course track.	SKIP	Content curriculum business, not a DAM feature.
Custom Virtual Builder (1 input -> XR module + textbook + video/podcast + quiz + exam)	This is effectively what B.2's LEARN + TRAIN items above already assemble from a single ingested document — kept as the platform's signature multi-output generation feature.	KEEP	Highest-leverage feature: one AI pipeline call producing several linked outputs from the same source asset.

Delivery channels & supporting systems

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
Virtual Campus (full course catalog, dual-diploma, AI-proctored exams)	Not built as an academic campus product.	SKIP	Institutional/diploma partnerships are a separate business line.
Enterprise Hub (white-labeled portal, SSO, compliance tracking)	The existing role-based dashboard is extended with basic org branding and the existing analytics/compliance tracking (already in baseline NFR-COMP).	ADAPT	Compliance tracking already exists in the baseline; white-label branding is a modest UI layer on top.
Consumer App (direct-to-individual app)	Not prioritized; the existing browser client plus WebAR already covers no-install consumer access.	SKIP	A dedicated consumer app is a distribution/marketing investment, not a DAM feature gap.
Marketplace (3rd-party creators publish/sell lessons)	A decentralized marketplace using the on-chain registry from Part A — creators list assets/lessons with licensing terms enforced by smart contract.	KEEP	This is where the blockchain layer pays off directly: licensing and royalty enforcement without a centralized storefront.
3D Asset Library	Already exists in the baseline (Sketchfab/Poly Pizza import + own uploads); extended with content-addressed IPFS storage from Part A.	KEEP	Core to VOID·SPACE's identity — already present, now provenance-backed.
AI Content Engine	Already exists (Claude API); extended output schema for lesson/quiz generation as described above.	KEEP	Already in baseline; this is scope extension, not new infrastructure.
AI Mentor/Tutor (in-lesson chat)	A scoped Claude-powered chat widget inside the lesson viewer, answering questions about the currently-viewed asset/lesson content only.	KEEP	Small, well-bounded addition to the existing viewer; reuses the existing Claude integration.
IoT Integration (live sensor/equipment data in simulations)	Not built.	SKIP	No current ingestion path for live equipment telemetry; large scope addition with no DAM overlap.

EON Feature / Product	What VOID·SPACE Would Build	Verdict	Reasoning
Multiplayer/Spatial Layer (shared real-time 3D space)	Not built in v1; noted as a future-phase item if Sentient Worlds-style features are revisited.	**SKIP**	Real-time networking infrastructure is a significant standalone engineering investment.
Analytics & Compliance Tracking	Already exists in the baseline (NFR-COMP); extended with on-chain verified-skill completion records.	**KEEP**	Already specified in VS-SRS-1.0 §6.7; blockchain adds tamper-evidence.

B.3 Proposed Architecture

The proposed platform keeps VOID·SPACE's existing ingestion -> AI enrichment -> review pipeline as its spine, and adds an XR lesson generation stage, a simulation/assessment stage, and routes verified outcomes into the blockchain-backed DAM and marketplace from Part A. 3D content creation tooling (Blender, Unity) sits alongside as the free, external toolchain feeding the asset library.

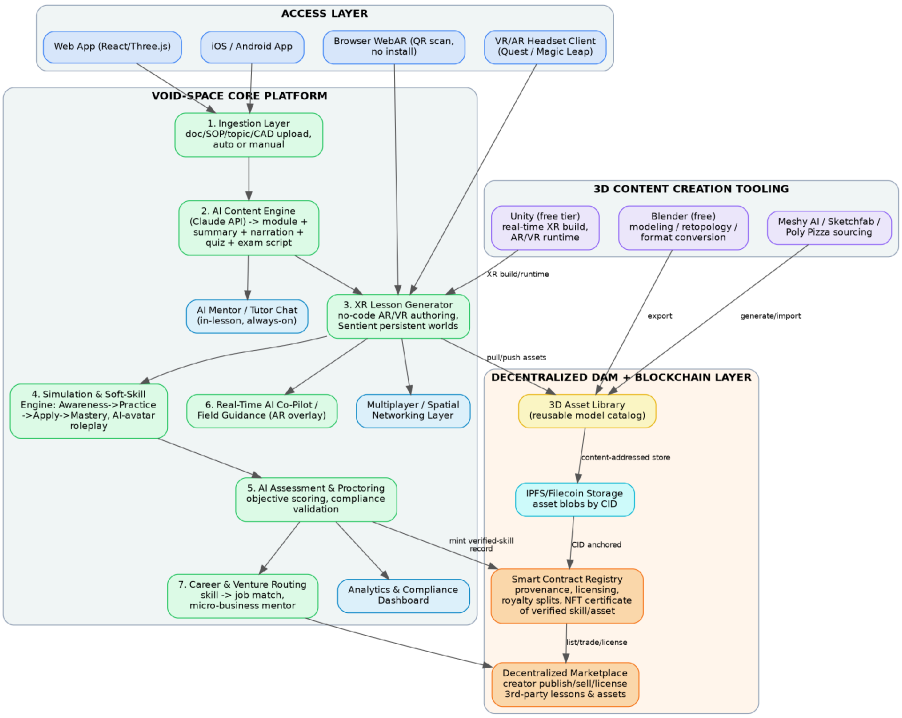


Figure B-1 — Proposed VOID·SPACE architecture: access layer, core platform, decentralized DAM/blockchain layer, and 3D tooling

B.4 Platform Lifecycle Loop

Reframing EON Reality's Learn -> Train -> Perform -> Achieve loop into VOID·SPACE's own DAM-centric terms:

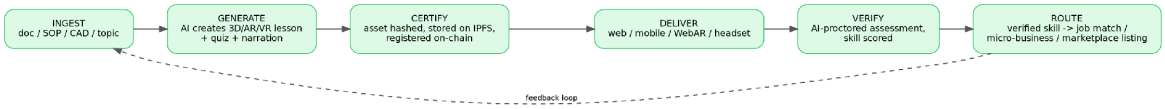


Figure B-2 — Ingest → Generate → Certify → Deliver → Verify → Route

- Ingest — a document, SOP, topic, or existing 3D/CAD asset enters the system (same entry point as the existing baseline upload flow).
- Generate — the AI content engine (Claude API, already in baseline) produces a 3D/AR lesson, summary, narration, and quiz from that input in one pass.
- Certify — on approval, the asset/lesson is hashed, pinned to IPFS, and registered on-chain (Part A's blockchain layer).
- Deliver — served through whichever channel fits: browser, mobile, WebAR (QR scan, no install), or VR/AR headset.
- Verify — an AI-assisted assessment scores the learner/user against the lesson content.

- Route — the verified outcome becomes a portable, on-chain credential and/or a marketplace listing, feeding back into what gets ingested next.

B.5 Blockchain-Native Additions Specific to the Proposed Platform

- NFT-style skill/asset certificates — a non-transferable (or transferable, per policy) token minted on successful assessment or asset approval, referencing the on-chain registry record from Part A.
- Smart-contract royalty splits — when a marketplace listing is licensed or sold, the contract automatically distributes proceeds between original Creator, remixers (if any), and the platform, without manual invoicing.
- Decentralized marketplace listing — any approved asset or lesson can be listed for license/sale directly from its on-chain registry entry; license terms are enforced by contract rather than a trust-based terms-of-service page.
- Optional DAO-style governance (future phase) — for community curation of the shared 3D asset library, flagged here as a later-phase idea rather than an MVP requirement.

B.6 3D Asset Design & Creation Tooling

Feature lists above assume assets already exist in the library or come from AI generation (Meshy AI) or 3rd-party import (Sketchfab/Poly Pizza). For assets that need to be modeled or built into an interactive XR experience by hand, two free tools cover the pipeline end-to-end:

Tool	Role in the pipeline	Cost
Blender	3D modeling, retopology/optimization for XR polycount budgets, and format conversion (already referenced in the existing SRS as the headless conversion tool — same tool, now also used for hands-on authoring).	Free, open-source
Unity (Personal/free tier)	Real-time XR build target — takes a modeled/optimized asset and turns it into an interactive AR/VR/WebAR experience or a simple training "game" (e.g. a scored interaction, a walkthrough)	Free tier available; paid tiers only needed at higher revenue thresholds

Workflow: model or source the asset (Blender or 3rd-party import) -> optimize polycount for XR (Blender) -> upload to VOID·SPACE (existing baseline flow) -> AI enrichment + review (existing) -> on approval, pin to IPFS and register on-chain (Part A) -> optionally package into an interactive Unity build for the lesson/training experience (Part B).

B.7 Delivery Channels & Device Support (Adapted)

Channel/Device	Status in proposed platform
Browser (WebGL, existing)	Already in baseline — unchanged.
Browser WebAR (QR scan, no install)	New — lets a lesson be viewed in AR directly from a phone camera without an app; low-cost addition on top of the existing WebGL viewer.
Native iOS/Android app	Not prioritized for MVP — see B.2 (SKIP), revisit only if user demand is confirmed.
VR headset (e.g. Meta Quest)	Optional Phase 2 target if Unity-built lessons need immersive delivery.
AR headset (e.g. Magic Leap)	Optional Phase 3 target — niche hardware, lower priority than WebAR.

B.8 Build Checklist (Proposed Platform Scope)

- [Existing] Content ingestion — document/topic/SOP/3D-asset upload.
- [Existing] AI generation pipeline — extend current Claude API schema to also emit lesson outline + quiz, not just tags/description.
- [Existing] Cross-platform viewer — WebGL viewer already present; add WebAR rendering mode.
- [New] Assessment engine — score a learner against generated quiz content; write result as a verified-skill record.
- [Existing] AI mentor/chat layer — scoped Claude-powered Q&A inside the lesson/asset viewer.
- [Existing] Admin/analytics dashboard — already in baseline NFR-COMP; extend with on-chain record status.
- [Existing] Content library — already in baseline; add IPFS content-addressing (Part A).
- [New] Blockchain registry & marketplace — smart contract for provenance/licensing/royalties; marketplace listing UI.
- [New] 3D tooling workflow docs — Blender export presets and Unity project template for turning approved assets into interactive builds.

B.9 Suggested Phased Roadmap

Phase	Scope
Phase 0 — Baseline	Existing VOID·SPACE SRS scope (VS-SRS-1.0) unchanged: auth, upload, review, preview, 3rd-party import, AI tagging, XR publish, API.
Phase 1 — Blockchain by default	IPFS pinning + on-chain registry (Part A.5) added to the baseline as a non-optional layer; wallet connect for signing.
Phase 2 — AI lesson generation	Extend the AI content engine schema to emit lesson outline, narration script, and quiz from the same ingested document (B.2 Custom Virtual Builder-equivalent).
Phase 3 — Assessment & credentials	Scored quiz/assessment flow; verified-skill record minted to the on-chain registry; portable credential export.
Phase 4 — Marketplace	Licensing/royalty smart contracts; marketplace listing and discovery UI.
Phase 5 — Delivery expansion (optional)	WebAR rendering; Unity-built interactive experiences; VR headset target if justified by demand.

Appendix — Quick Comparison

EON Reality vs. Existing VOID·SPACE vs. Proposed VOID·SPACE

Capability area	EON Reality	VOID·SPACE (existing)	VOID·SPACE (proposed)
Core identity	AI-powered XR training platform	3D Digital Asset Management (DAM) platform	DAM platform with AI lesson generation + blockchain provenance layered on top
Asset storage	Not disclosed (proprietary)	Server-side binary storage	IPFS/Filecoin content-addressed storage (added by default)
Provenance/licensing	Not disclosed	License metadata field only	On-chain smart-contract registry + enforced royalty splits
AI content generation	Full pipeline: 3D module + summary + narration + quiz + exam	Tags + description + quality flags only	Extended to also generate lesson outline + quiz (from the same Claude API integration)
Assessment	AI-proctored, compliance-grade	N/A (human review of assets only)	Lightweight AI-assisted quiz scoring, result written on-chain
Delivery	iOS/Android, desktop, VR/AR headsets, WebAR	Browser only	Browser + WebAR (MVP); headset delivery optional later phase
Marketplace	Yes (creator storefront)	None	Decentralized marketplace, contract-enforced licensing
Career/business routing	Career Compass, Venture Builder	None	Out of scope (SKIP) — portable credential export only

Document Source Traceability

- Part A content is derived from VOID·SPACE SRS v1.0 (VS-SRS-1.0), Sections 1-7, with the blockchain/IPFS layer added as new scope.
- Part B's EON Reality feature inventory is derived from the supplied “EON Reality — Platform & Feature Breakdown” reference document (13 features across 4 pillars, delivery channels, and supporting systems).
- This document is a planning/reference artifact intended to feed the next VOID·SPACE SRS revision (working title VS-SRS-2.0) — it is not itself a contractual SRS and does not use FR-/NFR- requirement IDs.